**Carisurg**

Assignment 18

Interim Documentation

**Due Date**: July 18th, 2026.

Prepared by **Josiah-John Green**

## Verdict

No. Neither Random Forest nor Gradient Boosting should replace the Week 6 logistic regression baseline: both trade away the one number that matters most in a triage system — recall on the most critical patients (ESI 1) — in exchange for a higher overall accuracy score, and that is not a trade worth making.

## Setup & Data Decisions

This is the same 55,121-encounter Yale EMMLC triage extract used since Week 5, the same top-10 feature shortlist carried over from the Week 5 memo (age, SpO2, chest pain, shortness of breath, suicidal ideation, alcohol intoxication, altered mental status, ambulance arrival, respiratory rate, heart rate), and the identical 80/20 stratified train/test split (random_state = 42) used in Week 6. No features changed this week — any difference in the numbers below comes from the model, not the data.

Two complex candidates were built and evaluated against the Week 6 baseline:

1. A **Random Forest** (300 trees, class_weight="balanced")
2. And a **Gradient Boosting** model (HistGradientBoostingClassifier, 300 boosting rounds).

A third variant — Random Forest **tuned specifically for macro recall** via RandomizedSearchCV — was added to test whether the weakest result below was a fixable hyperparameter problem or something more fundamental.

**Benchmark Table**

| **Model** | **Accuracy** | **Macro Recall** | **ESI-1 Recall** | **Weighted F1** | **Macro F1** | **Train (s)** | **Infer (ms/pred)** | **Interpretability** |
|---|---|---|---|---|---|---|---|---|
| *Logistic Regression (baseline)* | 0.254 | 0.417 | ***0.75*** | 0.327 | 0.214 | 0.337 | 0.0002 | High |
| *Random Forest* | 0.498 | 0.282 | ***0*** | 0.492 | 0.285 | 29.094 | 0.1218 | Medium |
| *Random Forest (tuned for recall)* | 0.397 | 0.35 | ***0*** | 0.423 | 0.268 | 146.826 | 0.0294 | Medium |
| *Gradient Boosting* | 0.341 | 0.405 | ***0.375*** | 0.389 | 0.253 | 0.657 | 0.0061 | Low |

## Appendices

### Supplementary Links

1. The GitHub link can be found here
2. Evidence for every figure above is included alongside this memo in the GitHub:
    a. **figs/01–07_*.png**
    b. **notebooks**/Week7_Optimization_Model

## Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

**Clinical & Vital Sign Terms**

| Term | Meaning |
|---|---|
| ED | Emergency Department |
| ESI | Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent) |
| SBP | Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading) |
| DBP | Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading) |
| MAP | Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as (SBP + 2×DBP) / 3 |
| GCS | Glasgow Coma Scale — score from 3–15, measuring level of consciousness |
| RR | Respiratory Rate — breaths per minute |

| | |
|---|---|
| FiO2 | Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%) |
| SpO2 | Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter |
| SBAR | Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication |

**Healthcare Systems & Infrastructure Terms**

| Term | Meaning |
|---|---|
| EHR | Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal |
| LMIC | Low- and Middle-Income Country — a World Bank income classification used in global health literature |
| SIDS | Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints |
| Boarding time | The gap between a disposition decision (e.g. “admit”) and the patient physically leaving the ED, usually due to no ward bed being available |

**AI / Machine Learning & Risk Terms**

| Term | Meaning |
|---|---|
| AI | Artificial Intelligence |
| ML | Machine Learning |
| AUROC / AUC | Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better |
| XGBoost | Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data |
| Rule-based classifier | A decision-making system using explicit if/then logic rather than a trained statistical model |
| PPV | Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue. |
| Automation bias | The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment |
| Distribution shift | When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on |
| Human-in-the-loop | A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect |

**Research & Documentation Terms**

| Term | Meaning |
|---|---|
| DOI | Digital Object Identifier — a permanent link format for academic papers |
| APA | American Psychological Association — the citation style used throughout this proposal (7th edition) |
| Scoping review | A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth |
| Systematic review | A more rigorous literature review following a defined search and inclusion protocol |